

2014 PJC JC 2 Prelim Exam Paper 1 Solution

1. Let \$a\$, \$b\$, and \$c\$ be the amount Angeline, Benjamin and Clement pay respectively.

For \$400 bill,

|            | Electricity | Water | Gas |
|------------|-------------|-------|-----|
| Amount(\$) | 200         | 160   | 40  |

Therefore,

$$0.4a + 0.5b + 0.7c = 200$$

$$0.6a + 0.3b + 0.2c = 160$$

$$0.2b + 0.1c = 40$$

$$\text{OR } a + b + c = 400$$

Solving with GC:  $a = 160$ ,  $b = 160$ ,  $c = 80$

i.e. Angeline pays \$160, Benjamin pays \$160, and Clement pays \$80

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$$t = \frac{\sqrt{100^2 + x^2}}{20} + \frac{1000 - x}{40} \Rightarrow \frac{dt}{dx} = \frac{1}{40} (100^2 + x^2)^{-\frac{1}{2}} (2x) - \frac{1}{40}$$

$$\frac{dt}{dx} = 0 \Rightarrow 2x = (100^2 + x^2)^{\frac{1}{2}}$$

$$\Rightarrow 4x^2 = x^2 + 100^2$$

$$\Rightarrow x = \frac{100}{\sqrt{3}} \quad (\because x > 0)$$

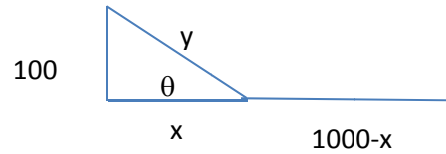
$$\therefore \text{required gradient of slope} = - \frac{100}{\frac{100}{\sqrt{3}}} = -\sqrt{3}$$

| $x$             | $\frac{100^-}{\sqrt{3}}$ | $\frac{100}{\sqrt{3}}$ | $\frac{100^+}{\sqrt{3}}$ |
|-----------------|--------------------------|------------------------|--------------------------|
| $\frac{dt}{dx}$ | -ve                      | 0                      | +ve                      |

$$\text{OR } \frac{d^2t}{dx^2} = \frac{\sqrt{100^2 + x^2} - \frac{x^2}{\sqrt{100^2 + x^2}}}{20^2 (100^2 + x^2)} = \frac{2x - \frac{x^2}{2x}}{20^2 (4x^2)} = \frac{3}{8x(20^2)} > 0 \text{ (min occurs)}$$

**Alternative solution:**

From diagram,  $x = \frac{100}{\tan \theta}$ ,  $y = \frac{100}{\sin \theta}$ ,



Let  $t$  be the total time taken.

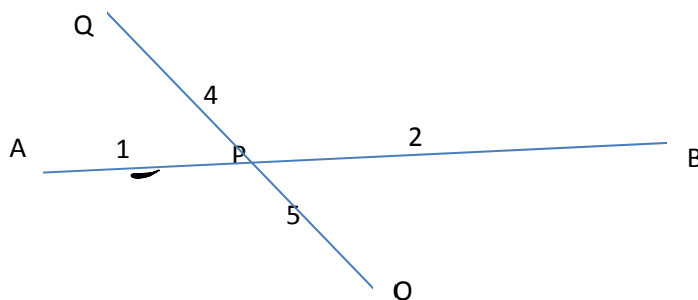
$$\begin{aligned}
 t &= \frac{y}{20} + \frac{1000-x}{40} \Rightarrow t = \frac{100/\sin \theta}{20} + \frac{1000-100/\tan \theta}{40} \\
 &\Rightarrow t = 25 + \frac{5}{\sin \theta} - \frac{5}{2 \tan \theta} \\
 &\Rightarrow \frac{dt}{d\theta} = -5(\sin \theta)^{-2} \cos \theta + \frac{5}{2} (\tan \theta)^{-2} \sec^2 \theta \\
 \frac{dt}{d\theta} &= 0 \Rightarrow \frac{5 \sec^2 \theta}{2 \tan^2 \theta} = \frac{5 \cos \theta}{\sin^2 \theta} \\
 &\Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}
 \end{aligned}$$

Use first derivative test:

| $\theta$             | $\frac{\pi}{3}^-$ | $\frac{\pi}{3}$ | $\frac{\pi}{3}^+$ |
|----------------------|-------------------|-----------------|-------------------|
| $\frac{dt}{d\theta}$ | -ve               | 0               | +ve               |

$\therefore$  required gradient of slope =  $-\tan \frac{\pi}{3} = -\sqrt{3}$

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(i) Using Ratio Theorem,

$$\overrightarrow{OP} = \frac{2\mathbf{a} + \mathbf{b}}{3}$$

$$\text{Then, } \overrightarrow{OQ} = \frac{9}{5} \overrightarrow{OP} = \frac{3}{5} (2\mathbf{a} + \mathbf{b})$$

$$\begin{aligned}
 \text{(ii)} \quad \overrightarrow{QC} &= \overrightarrow{OC} - \overrightarrow{OQ} \\
 &= 3\mathbf{a} - \frac{3}{5}(2\mathbf{a} + \mathbf{b}) \\
 &= \frac{3}{5}(3\mathbf{a} - \mathbf{b}) = \frac{3}{5}\overrightarrow{BC}
 \end{aligned}$$

Since  $\overrightarrow{QC} = \frac{3}{5}\overrightarrow{BC}$ , and  $C$  is a common point,  $Q$  lies on  $BC$ .

Since  $|\overrightarrow{BQ}| : |\overrightarrow{QC}| = 2 : 3 \Rightarrow BQ : QC = 2 : 3$

OR

$$\begin{aligned}
 \overrightarrow{BQ} &= \overrightarrow{OQ} - \overrightarrow{OB} \\
 &= \frac{3}{5}(2\mathbf{a} + \mathbf{b}) - \mathbf{b} \\
 &= \frac{2}{5}(3\mathbf{a} - \mathbf{b}) \\
 &= \frac{2}{5}\overrightarrow{BC}
 \end{aligned}$$

**Alternative Solution to show that  $Q$  lies on  $BC$ :**

Equation of line  $BC$ :

$$\mathbf{r} = \mathbf{b} + \lambda(3\mathbf{a} - \mathbf{b}) = 3\lambda\mathbf{a} + (1 - \lambda)\mathbf{b}$$

If  $Q$  lies on  $BC$ ,

$$\begin{aligned}
 \frac{3}{5}(2\mathbf{a} + \mathbf{b}) &= 3\lambda\mathbf{a} + (1 - \lambda)\mathbf{b} \\
 \frac{6}{5}\mathbf{a} + \frac{3}{5}\mathbf{b} &= 3\lambda\mathbf{a} + (1 - \lambda)\mathbf{b}
 \end{aligned}$$

$$\text{Therefore, } 3\lambda = \frac{6}{5} \Rightarrow \lambda = \frac{2}{5} \text{ and } 1 - \lambda = \frac{3}{5} \Rightarrow \lambda = \frac{2}{5}$$

Since  $\lambda$  is unique, point  $Q$  lies on  $BC$ .

Note:

Alternative equation of line  $BC$   
 $\mathbf{r} = 3\mathbf{a} + \lambda(3\mathbf{a} - \mathbf{b}) = (3 + 3\lambda)\mathbf{a} - \lambda\mathbf{b}$   
 may be used

(iii) Area of triangle  $ABC$

$$\begin{aligned}
 &= \frac{1}{2} |\overrightarrow{AP} \times \overrightarrow{AQ}| \\
 &= \frac{1}{2} \left| \frac{1}{3}(\mathbf{b} - \mathbf{a}) \times \frac{1}{5}(\mathbf{a} + 3\mathbf{b}) \right| \\
 &= \frac{1}{30} |(\mathbf{b} - \mathbf{a}) \times (\mathbf{a} + 3\mathbf{b})| \\
 &= \frac{1}{30} |\mathbf{b} \times \mathbf{a} - 3\mathbf{a} \times \mathbf{b}| \\
 &= \frac{1}{30} |-4\mathbf{a} \times \mathbf{b}| \\
 &= \frac{2}{15} |\mathbf{a} \times \mathbf{b}|
 \end{aligned}$$

Therefore,  $k = \frac{2}{15}$

Note:

Can also use other sides to cross such as  $\frac{1}{2} |\overrightarrow{AP} \times \overrightarrow{PQ}|$

4 (i)

$$u_2 - u_1 = \frac{2(1)^2 - 1}{4(1)^2 - 1} \Rightarrow u_2 = \frac{4}{3} = \frac{2^2}{3}$$

$$u_3 - u_2 = \frac{2(2)^2 - 1}{4(2)^2 - 1} \Rightarrow u_3 = \frac{27}{15} = \frac{9}{5} = \frac{3^2}{5}$$

$$u_4 - u_3 = \frac{2(3)^2 - 1}{4(3)^2 - 1} \Rightarrow u_4 = \frac{80}{35} = \frac{16}{7} = \frac{4^2}{7}$$

$$u_n = \frac{n^2}{2n-1}$$

(i) Let  $P_n$  be the statement

$$u_n = \frac{n^2}{2n-1} \quad \text{for } n = 1, 2, 3, \dots$$

When  $n = 1$ ,

LHS =  $u_1 = 1$  (given)

$$\text{RHS} = \frac{1^2}{2(1)-1} = 1$$

Thus  $P_1$  is true

Assume that  $P_k$  is true for some  $k$ , where  $k = 1, 2, 3, \dots$

$$u_k = \frac{k^2}{2k-1}$$

To prove that  $P_{k+1}$  is also true i.e.

$$u_{k+1} = \frac{(k+1)^2}{2k+1}$$

$$\text{LHS} = u_{k+1}$$

$$\begin{aligned} &= u_k + \frac{2k^2-1}{4k^2-1} \\ &= \frac{k^2}{2k-1} + \frac{2k^2-1}{4k^2-1} \\ &= \frac{k^2(2k+1) + 2k^2-1}{(2k-1)(2k+1)} \\ &= \frac{2k^3 + 3k^2 - 1}{(2k-1)(2k+1)} \\ &= \frac{(2k-1)(k^2 + 2k + 1)}{(2k-1)(2k+1)} \\ &= \frac{(k+1)^2}{2k+1} \\ &= \text{RHS} \end{aligned}$$

Therefore  $P_{k+1}$  is also true

Since  $P_1$  is true and  $P_k$  is true  $\Rightarrow P_{k+1}$  is also true, then by mathematical induction

$$u_n = \frac{n^2}{2n-1} \text{ for } n=1, 2, 3, \dots$$

(ii) When  $n \rightarrow \infty$ ,  $u_n = \frac{1}{2}n + \frac{1}{4} + \frac{1}{4(2n-1)} \rightarrow \infty$ , therefore the sequence is not convergent.

$$5 \text{ (i)} \quad \frac{d^2z}{dy^2} = \frac{-1}{(2+y)^2} \quad \Rightarrow \quad \frac{dz}{dy} = \frac{1}{2+y} + C$$

$$\text{When } y = -1, \frac{dz}{dy} = 1 \text{ and } z = 0 \Rightarrow 1 = 1 + C \Rightarrow C = 0$$

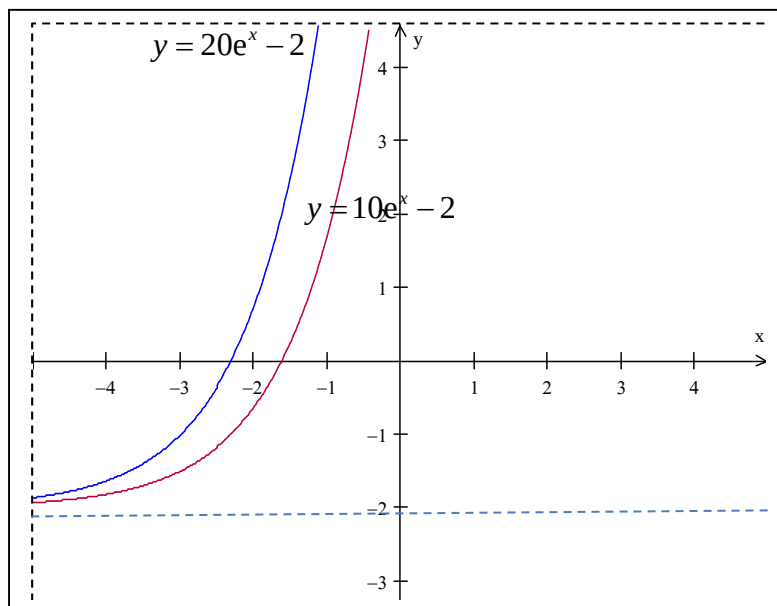
$$\frac{dz}{dy} = \frac{1}{2+y} \Rightarrow z = \ln|2+y| + D$$

$$0 = \ln 1 + 0 + D \Rightarrow D = 0 \Rightarrow \quad z = \ln|2+y| = \ln(2+y) \quad (\because 2+y > 0)$$

(ii)

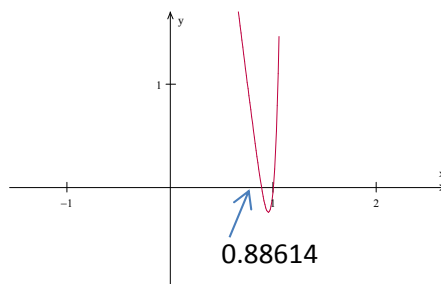
$$\begin{aligned}\frac{dy}{dx} &= e^z & \Rightarrow \frac{dy}{dx} &= e^{\ln(2+y)} = 2+y \\ & & \Rightarrow \int \frac{1}{2+y} dy &= \int 1 dx \\ & & \Rightarrow \ln(2+y) &= x + K \\ & \Rightarrow y &= Qe^x - 2, & Q = e^K\end{aligned}$$

(iii)

Non-lineareg.  $y = 10e^x - 2$ ,  $y = 20e^x - 2$ 

6 (i)

$$\begin{aligned}
 1000 &= \frac{125(1-r^{20})}{1-r} \\
 \Rightarrow 8-8r &= 1-r^{20} \\
 \Rightarrow r^{20}-8r+7 &= 0 \\
 \Rightarrow r &\approx 0.88614 \quad (\text{by GC})
 \end{aligned}$$



$$\text{Total petrol used from after 20th kilometre} \approx \frac{125}{1-0.88614} - 1000 \approx 97.8 < 98 \quad (\text{shown})$$

(ii)

$$\begin{aligned}
 T_5 &= 1020 \\
 \Rightarrow 1000 + 4d &= 1020 \\
 \Rightarrow d &= 5
 \end{aligned}$$

(iii)

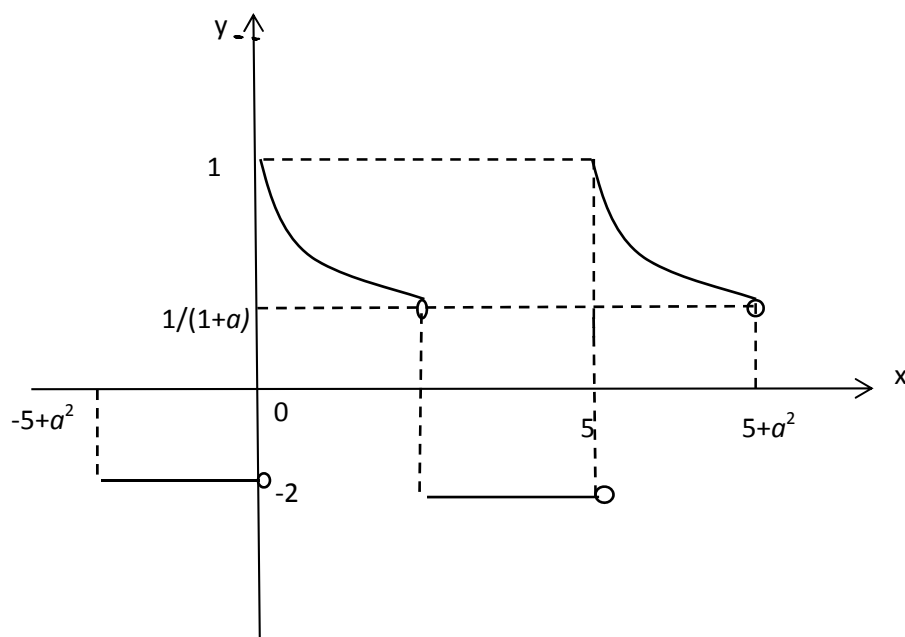
For car A, petrol used in 24th km  $= T_{24} \approx 125(0.88614^{23}) \approx 7.7534$  millilitres

For car B, petrol used in 24th km  $= d = 5$  millilitres (by (ii))

$\therefore$  car B has better fuel efficiency at the 24th km.

$$7 \quad \text{(i)} \quad f(100) = f(0) = 1$$

(ii)



(iii)

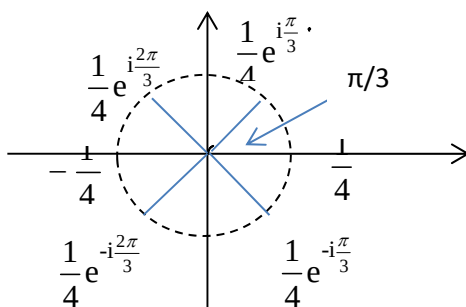
$$\begin{aligned}
 & \int_0^{a^2} \frac{1}{1+\sqrt{x}} dx \\
 &= \int_1^{1+a} \frac{1}{u} 2(u-1) du \\
 &= 2 \int_1^{1+a} 1 - \frac{1}{u} du = 2 \left[ u - \ln|u| \right]_1^{1+a} = 2(1+a - \ln|1+a| - 1 + 0) = 2(a - \ln(1+a))
 \end{aligned}$$

$$\begin{aligned}
 u &= 1 + \sqrt{x} \\
 \frac{du}{dx} &= \frac{1}{2\sqrt{x}} = \frac{1}{2(u-1)}
 \end{aligned}$$

8 (i)  $(1 - \sqrt{3}i)^{-12} = \left( 2e^{-i\frac{\pi}{3}} \right)^{-12} = 2^{-12} e^{i4\pi} = \frac{1}{4096} \quad (\text{Shown})$

(ii)

$$\begin{aligned}
 z^6 (1 - \sqrt{3}i)^{12} &= 1 \quad \Rightarrow z^6 = (1 - \sqrt{3}i)^{-12} = \frac{1}{4096} = \frac{1}{4096} e^{i0} \\
 \Rightarrow z^6 &= \frac{1}{4096} e^{i\pi(2n)/6}, n = 0, \pm 1, \pm 2, 3 \\
 \Rightarrow z &= \frac{1}{4} e^{i\frac{(n)\pi}{3}}, n = 0, \pm 1, \pm 2, 3 \\
 \Rightarrow z &= \frac{1}{4} e^{i\frac{\pi}{3}}, \frac{1}{4} e^{-i\frac{\pi}{3}}, \frac{1}{4} e^{i\frac{2\pi}{3}}, \frac{1}{4} e^{-i\frac{2\pi}{3}}, \frac{1}{4}, -\frac{1}{4}
 \end{aligned}$$



(iii)

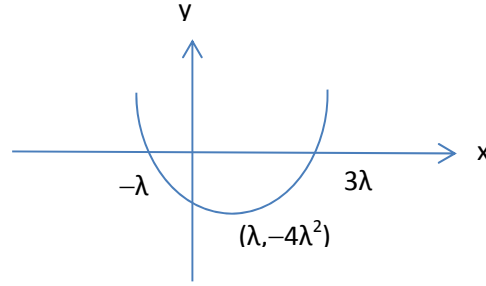
$$\begin{aligned}
 (z - re^{i\theta})(z - re^{-i\theta}) &= z^2 - r(e^{i\theta} + e^{-i\theta})z + r^2 \\
 &= z^2 - r(2\cos\theta)z + r^2
 \end{aligned}$$

(iv)

$$\begin{aligned}
z^6(1-\sqrt{3}i)^{12}-1 &= 4096\left(z-\frac{1}{4}e^{i\frac{\pi}{3}}\right)\left(z-\frac{1}{4}e^{-i\frac{\pi}{3}}\right)\left(z-\frac{1}{4}e^{\frac{2\pi}{3}i}\right)\left(z-\frac{1}{4}e^{-i\frac{2\pi}{3}}\right)\left(z-\frac{1}{4}\right)\left(z+\frac{1}{4}\right) \\
&= 4096\left(z^2-\frac{1}{2}z\cos\frac{\pi}{3}+\frac{1}{16}\right)\left(z^2-\frac{1}{2}z\cos\frac{2\pi}{3}+\frac{1}{16}\right)\left(z^2-\frac{1}{16}\right) \\
&= (16z^2-4z+1)(16z^2+4z+1)(16z^2-1)
\end{aligned}$$

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(i)

(ii)  $k = \lambda$ .

(iii)

$$\begin{aligned}
\text{Let } y &= x^2 - 2x - 3, \quad x \geq 1, \\
\Rightarrow x^2 - 2x - 3 - y &= 0 \\
\Rightarrow x &= \frac{2 \pm \sqrt{4 - 4(-3 - y)}}{2} = 1 \pm \sqrt{y + 4} \\
\Rightarrow x &= 1 + \sqrt{y + 4} \quad (\text{since } x \geq 1) \quad \Rightarrow f^{-1}(x) = 1 + \sqrt{x + 4}, \quad x \geq -4
\end{aligned}$$

(iv)

$$f^2(x) = x \quad \Rightarrow f(x) = f^{-1}(x)$$

Since  $y = f(x)$ ,  $y = f^{-1}(x)$  and  $y = x$  intersect at the same point.

$$\begin{aligned}
\therefore f(x) = x &\Rightarrow x^2 - 2x - 3 = x \\
&\Rightarrow x^2 - 3x - 3 = 0 \\
&\Rightarrow x = \frac{3 \pm \sqrt{21}}{2} \Rightarrow x = \frac{3 + \sqrt{21}}{2} \quad (\because x \geq 1)
\end{aligned}$$

- (v)  $y = x^2 - 2x - 3$   
 $\Rightarrow y = (x-1)^2 - 4$   
 $\therefore$  (1) Translate  $y = x^2$  in the direction of  $x$ -axis by 1 unit  
 (2) Translate  $y = (x-1)^2$  in the direction of  $y$ -axis by  $-4$  units

- 10 (i) Since there is an asymptote at  $x = 2$ ,  $2^2 + r = 0 \Rightarrow r = -4$

Therefore equation of C:  $y = \frac{px+q}{x^2-4}$

$\left(4, \frac{1}{4}\right)$  is a point on C:

$$\frac{1}{4} = \frac{4p+q}{12} \Rightarrow q = 3-4p \quad \dots(1)$$

$$\frac{dy}{dx} = \frac{p(x^2-4) - 2x(px+q)}{(x^2-4)^2} = \frac{-px^2 - 4p - 2qx}{(x^2-4)^2}$$

$\left(4, \frac{1}{4}\right)$  is a turning point on C:

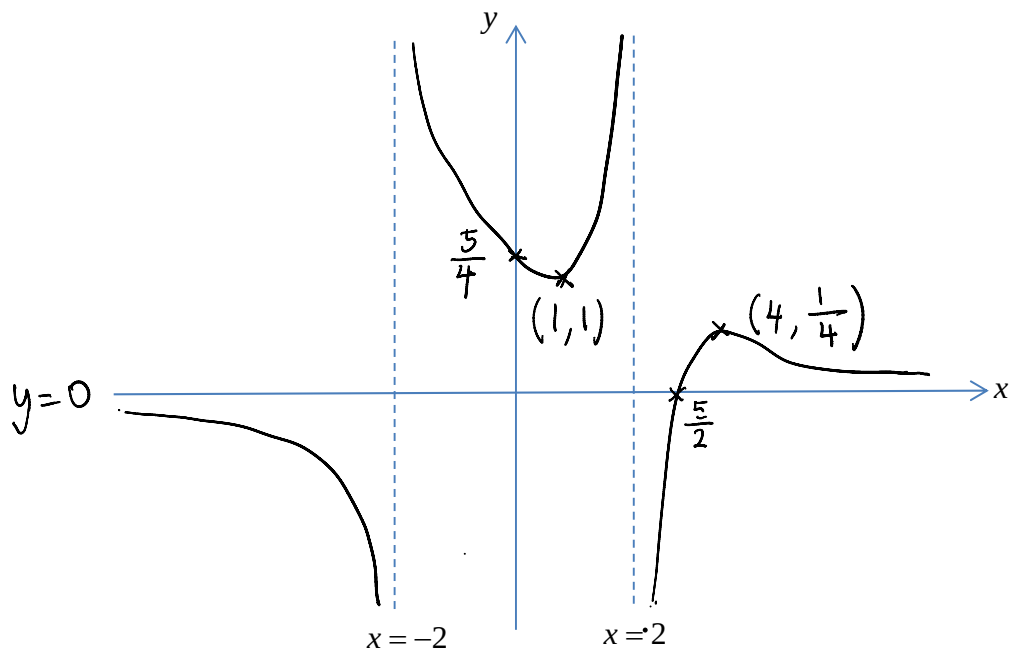
$$-16p - 4p - 8q = 0$$

$$5p = -2q \dots(2)$$

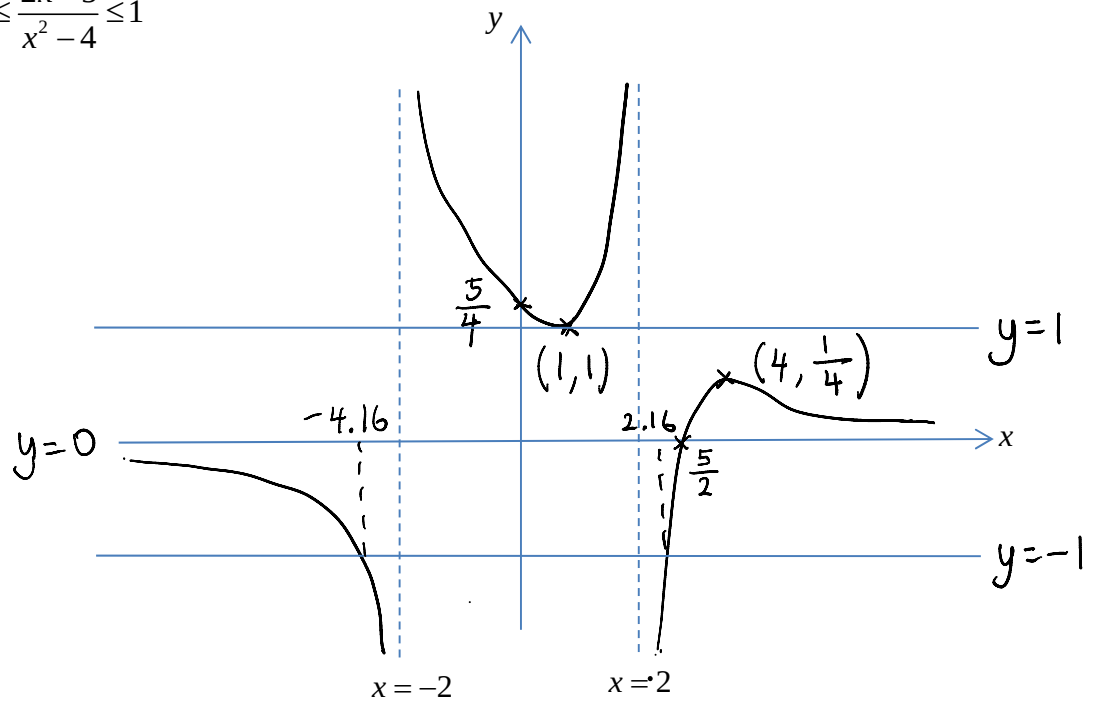
Solving (1) and (2),  $p = 2, q = -5$

Therefore equation of C:  $y = \frac{2x-5}{x^2-4}$

(i)

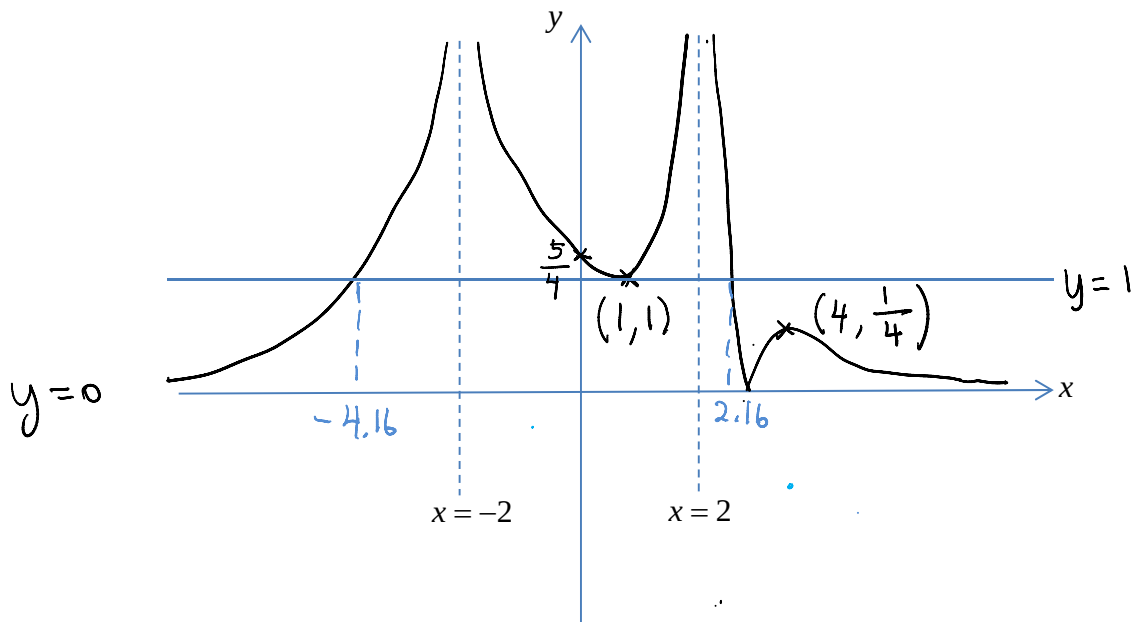


(ii)  $-1 \leq \frac{2x-5}{x^2-4} \leq 1$



$$x \leq -4.16 \quad \text{or} \quad x = 1 \quad \text{or} \quad x \geq 2.16$$

Alternative Solution:



$$x \leq -4.16 \quad \text{or} \quad x = 1 \quad \text{or} \quad x \geq 2.16$$

11 (i) Point  $A(2, 0)$

$$\frac{dx}{dt} = \frac{4}{(1-t)^3} \text{ and } \frac{dy}{dt} = \frac{1}{(1-t)^2}$$

$$\frac{dy}{dx} = \frac{\frac{1}{(1-t)^2}}{\frac{4}{(1-t)^3}} = \frac{1}{4}(1-t)$$

$$\text{Equation of normal: } y - \frac{t}{1-t} = -\frac{4}{1-t} \left( x - \frac{2}{(1-t)^2} \right)$$

Since the normal at  $P$  and  $Q$  pass through  $A$ ,

$$0 - \frac{t}{1-t} = -\frac{4}{1-t} \left( 2 - \frac{2}{(1-t)^2} \right)$$

$$\frac{8}{(1-t)^3} = \frac{8-t}{1-t}$$

Simplifying,

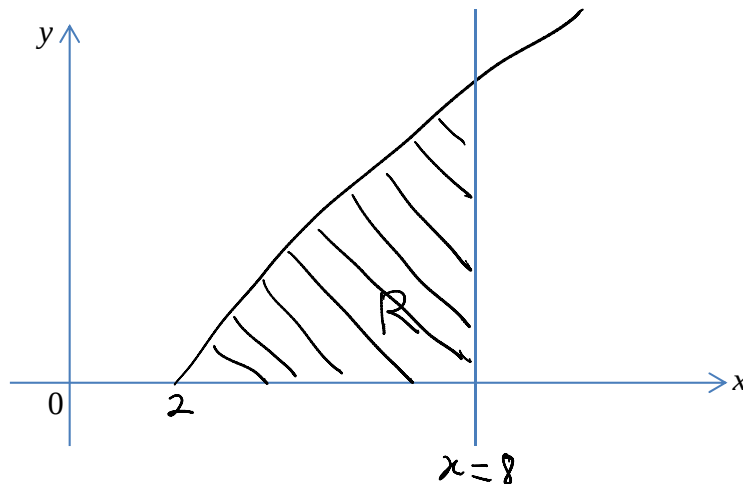
$$t^3 - 10t^2 + 17t = 0$$

$$t(t^2 - 10t + 17) = 0$$

$$t = 0, \quad t^2 - 10t + 17 = 0 \Rightarrow t = 5 \pm 2\sqrt{2}$$

$$\text{Therefore, } p = 5 - 2\sqrt{2}, \quad q = 5 + 2\sqrt{2}$$

(ii)



When  $x = 8$ ,

$$\frac{2}{(1-t)^2} = 8 \Rightarrow t = \frac{1}{2} \text{ or } \frac{3}{2} \text{ (rejected since } y \geq 0 \text{)}$$

Area of region  $R$

$$\begin{aligned}
 &= \int_2^8 y \, dx \\
 &= \int_0^{\frac{1}{2}} \frac{t}{1-t} \left( \frac{4}{(1-t)^3} \right) dt \\
 &= \int_0^{\frac{1}{2}} \frac{4t}{(1-t)^4} dt \\
 &= \frac{10}{3} \quad (\text{using GC})
 \end{aligned}$$

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$$(i) \quad \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} \times \begin{pmatrix} 0 \\ -4 \\ 5 \end{pmatrix} = \begin{pmatrix} 12 \\ -10 \\ -8 \end{pmatrix} = 2 \begin{pmatrix} 6 \\ -5 \\ -4 \end{pmatrix} \Rightarrow \text{Normal to } p_1 = \begin{pmatrix} 6 \\ -5 \\ -4 \end{pmatrix}$$

$$\text{Vector equation of } p_1 \text{ is } \mathbf{r} \cdot \begin{pmatrix} 6 \\ -5 \\ -4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 7 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ -5 \\ -4 \end{pmatrix} = -32$$

Hence the Cartesian equation of  $p_1$  and  $p_2$  are

$$6x - 5y - 4z = -32$$

$$5x - y + 3z = 24$$

Since the lines intersect, using GC,

$$\text{Vector equation of line } l \text{ is } \mathbf{r} = \begin{pmatrix} 8 \\ 16 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$$

$$(ii) \quad \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix} - \begin{pmatrix} 8 \\ 16 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 \\ -20 \\ 1 \end{pmatrix}$$

Therefore 2 direction vectors parallel to  $p_3$  are  $\begin{pmatrix} -5 \\ -20 \\ 1 \end{pmatrix}$  and  $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

$$\begin{pmatrix} -5 \\ -20 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 18 \\ -4 \\ 10 \end{pmatrix} = 2 \begin{pmatrix} 9 \\ -2 \\ 5 \end{pmatrix}$$

Normal to  $p_3$  is  $\begin{pmatrix} 9 \\ -2 \\ 5 \end{pmatrix}$

Therefore, vector equation of plane  $p_3$ :  $\mathbf{r} \bullet \begin{pmatrix} 9 \\ -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix} \bullet \begin{pmatrix} 9 \\ -2 \\ 5 \end{pmatrix} = 40$

Thus, Cartesian equation of plane  $p_3$ :  $9x - 2y + 5z = 40$

- (iii) The systems of equations correspond to the equations of planes  $p_1$ ,  $p_2$  and  $p_3$ .  
 Since all 3 plane contain or intersect at line  $l$ , the system of equations has an infinite number of solutions.